

GANDER, NL, CanadaWMO: **717420**Lat: **48.946N**Lon: **54.567W**Elev: **151**StdP: **99.52**Time Zone: **-3.50 (NAF)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-17.3	-15.3	-23.0	0.5	-16.2	-20.6	0.6	-14.6	17.7	1.7	15.4	-1.7	6.2	280	0.767

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.7	26.4	18.7	24.9	18.0	23.3	17.3	20.5	24.3	19.5	22.9	18.5	21.7	5.4	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.2	14.2	22.6	18.2	13.3	21.5	17.2	12.5	20.5	59.9	24.4	56.4	22.9	53.1	21.8	23.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.3	11.7	10.4	-20.0	29.8	2.5	1.2	-21.8	30.7	-23.3	31.4	-24.7	32.1	-26.5	32.9		
(5)			-20.5	22.5	2.5	0.7	-22.3	23.0	-23.7	23.4	-25.1	23.8	-26.9	24.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	4.8	-6.1	-6.3	-3.4	1.6	6.8	11.8	17.1	16.8	12.4	6.9	2.0	-2.9
	DBStd	9.12	4.69	5.00	4.34	3.59	3.93	4.45	3.66	3.25	3.84	3.72	4.06	4.42
	HDD10.0	2547	498	457	417	253	117	31	2	1	18	113	241	401
	HDD18.3	4999	756	690	675	502	357	201	66	68	181	356	489	659
	CDD10.0	643	0	0	0	1	18	84	221	212	91	16	2	0
	CDD18.3	54	0	0	0	0	0	4	27	20	3	0	0	0
	CDH23.3	335	0	0	0	0	6	39	177	98	16	0	0	0
	CDH26.7	39	0	0	0	0	1	4	23	10	1	0	0	0
Wind	WSAvg	5.5	6.3	6.3	6.1	5.8	5.2	4.9	4.6	4.4	5.0	5.3	5.7	6.1
Precipitation	PrecAvg	1239	114	105	116	112	86	85	83	94	94	112	117	115
	PrecMax	1668	226	276	269	279	279	193	188	190	226	220	205	195
	PrecMin	692	37	35	22	41	0	25	11	13	0	20	6	41
	PrecStd	181	45	45	54	54	48	39	37	46	39	48	43	37
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.0	8.6	8.7	15.6	23.2	26.7	29.1	27.9	25.5	19.4	15.2	10.7
		MCWB	9.4	7.4	6.2	10.3	14.7	18.0	20.3	20.3	18.8	16.3	13.6	10.1
	2%	DB	5.5	5.5	6.0	12.0	19.5	24.3	27.0	25.7	22.3	16.9	13.2	7.9
		MCWB	5.0	4.8	4.4	8.8	12.9	16.5	19.1	18.6	17.5	14.6	12.3	7.3
	5%	DB	2.8	2.7	4.0	9.5	16.6	22.1	25.4	24.2	20.4	14.9	11.0	5.1
		MCWB	2.2	1.9	2.4	6.5	11.3	15.7	18.4	18.0	16.5	12.8	10.2	4.6
	10%	DB	0.5	0.6	2.3	7.3	14.1	19.8	23.6	22.7	18.7	12.9	8.6	3.0
		MCWB	-0.1	-0.1	0.9	4.6	9.8	14.6	17.8	17.3	15.4	10.9	7.5	2.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.3	8.0	7.4	12.0	16.3	20.3	22.1	21.5	20.2	17.3	14.0	10.2
		MCDB	9.9	8.5	8.7	13.4	20.4	24.4	26.7	25.7	23.1	18.5	14.7	10.6
	2%	WB	5.1	4.8	4.6	9.2	14.1	18.3	20.8	20.2	18.7	15.1	12.3	7.4
		MCDB	5.5	5.4	5.5	11.6	18.1	21.9	24.9	23.7	21.2	16.3	13.1	7.8
	5%	WB	2.3	2.1	2.6	6.9	12.1	16.6	19.7	19.2	17.3	13.1	10.2	4.7
		MCDB	2.8	2.6	3.6	9.0	15.3	20.5	23.4	22.6	19.6	14.6	10.9	5.0
	10%	WB	0.0	-0.1	1.1	5.0	10.3	15.1	18.7	18.3	16.0	11.1	7.6	2.5
		MCDB	0.3	0.4	2.1	6.8	13.8	19.0	22.1	21.5	18.2	12.6	8.5	3.0
Mean Daily Temperature Range	5% DB	MDBR	7.0	7.4	7.2	7.3	9.4	10.1	9.7	8.9	8.3	6.6	6.0	5.8
		MCDBR	10.2	10.1	9.2	11.1	14.6	13.7	12.2	11.1	10.6	9.5	9.4	9.0
		MCWBR	10.5	9.9	8.0	7.9	8.9	7.9	6.5	6.0	6.7	7.7	9.2	9.2
	5% WB	MCDBR	10.2	10.0	8.6	10.4	13.1	11.9	10.7	9.8	9.5	9.2	9.4	9.0
		MCWBR	10.5	10.1	8.1	8.4	8.9	7.7	6.5	6.2	7.1	8.3	9.4	9.3
Clear-Sky Solar Irradiance	taub	0.304	0.302	0.332	0.351	0.366	0.388	0.407	0.388	0.355	0.332	0.309	0.302	
	taud	2.274	2.342	2.324	2.385	2.404	2.358	2.347	2.397	2.482	2.511	2.496	2.325	
	Ebn at Noon	740	850	879	898	895	872	849	850	848	810	752	692	
	Edh at Noon	73	87	105	110	112	118	118	108	90	74	60	62	
All-Sky Solar Radiation	RadAvg	0.96	1.73	2.86	3.95	4.68	5.01	5.19	4.28	3.27	2.00	1.09	0.71	
	RadStd	0.09	0.21	0.26	0.45	0.38	0.57	0.43	0.32	0.23	0.21	0.12	0.07	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	+0.43	+0.71	N/A	N/A	N/A
(47) Regional (12 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	+0.53	N/A	N/A	N/A

Nomenclature: See separate page